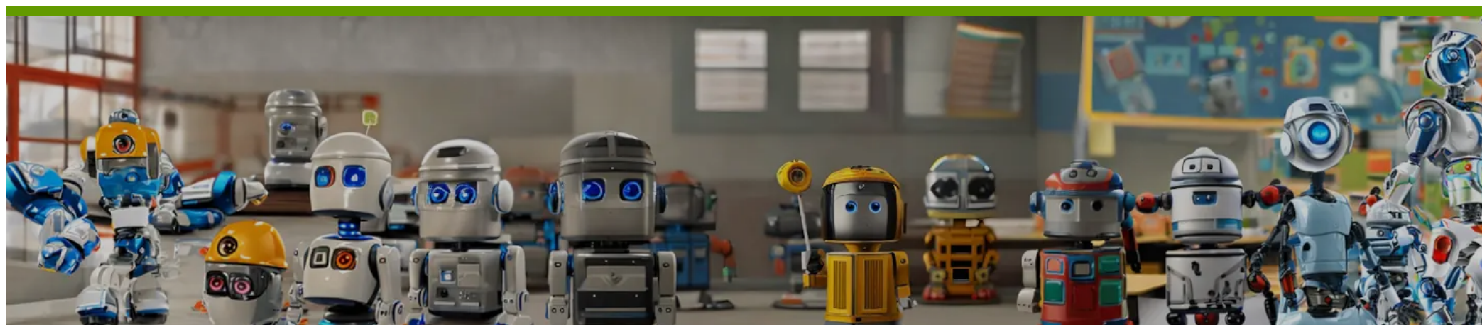




Machine Learning Bits > Non-negative Matrix Factorization

Non-negative Matrix Factorization

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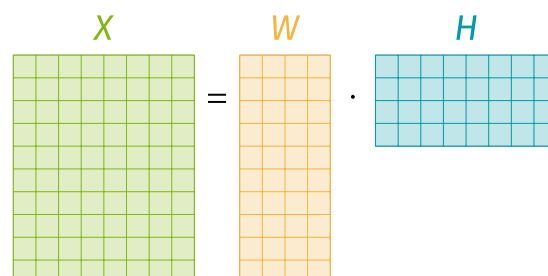
Non-negative Matrix Factorization (NMF, NNMF) [LeSe99]

Given a data matrix $X \in \mathbb{R}^{m \times n} \geq 0$, and the parameter k with $1 \leq k \leq \min\{m, n\}$, find two matrixes

$$\begin{aligned} W &\in \mathbb{R}^{m \times k} \geq 0 \\ H &\in \mathbb{R}^{k \times n} \geq 0 \end{aligned}$$

such that

$$X \approx WH$$



Variant 1: Squared error – minimize

$$\|X - WH\|^2 = \sum_i \sum_j (X_{ij} - (WH)_{ij})^2$$

Variant 2: Divergence – minimize the log-likelihood of $X_{ij} \sim \text{Poisson}(WH)_{ij}$

$$L(W, H) = - \sum_i \sum_j X_{ij} \log (WH)_{ij} - (WH)_{ij}$$

NMF: Multiplicative Update for Squared Error

?

$$\|X - WH\|^2 = \sum_i \sum_j (X_{ij} - (WH)_{ij})^2$$

1. initialize with random non-negative values (later: with better heuristics)
2. maximize W :

$$w_{ik} \leftarrow w_{ik} \frac{(XH^T)_{ik}}{(WHH^T)_{ik}}$$

3. maximize H :

$$h_{kj} \leftarrow h_{kj} \frac{(W^T X)_{kj}}{(W^T WH)_{kj}}$$

4. repeat 2-3 until the improvement of $\|X - WH\|^2$ is below a threshold

May get stuck in a local fixpoint or saddlepoint – does not guarantee to find the optimum.

Note: we need $w_{ij} > 0$, $h_{ij} > 0$, so it is common to add a small constant $\epsilon = 10^{-16}$ to all values.

NMF: Multiplicative Update for KL-Divergence

Gradient descent method for KL-Divergence [LeSe00; LeSe99]:

$$L(W, H) = \sum_i \sum_j X_{ij} \log (WH)_{ij} - (WH)_{ij}$$

1. initialize randomly (or with better heuristics)
2. maximize W :

$$w_{ik} \leftarrow w_{ik} \frac{\sum_{j=1}^m h_{kj} \frac{x_{ij}}{(WH)_{ij}}}{\sum_{j=1}^m h_{kj}}$$

3. maximize H :

$$h_{kj} \leftarrow h_{kj} \frac{\sum_{i=1}^n w_{ik} \frac{x_{ij}}{(WH)_{ij}}}{\sum_{i=1}^n w_{ik}}$$

4. repeat 2-3 until the improvement of $L(W, H)$ is below a threshold

May get stuck in a local fixpoint or saddlepoint – does not guarantee to find the optimum.

Note: we need $w_{ij} > 0$, $h_{ij} > 0$, so it is common to add a small constant $\epsilon = 10^{-16}$ to all values.

Other Algorithms for NMF



Many algorithm variants have been developed:

- ▶ Alternating Least Squares [PaTa94]
- ▶ Projected-gradient ALS [Lin07]
- ▶ Quasi-Newton optimization [ZdCi06]
- ▶ Hierarchical Alternating Least Squares [CiPh09]
- ▶ Initialize with spherical- k -means
- ▶ Initialize with SVD [BBLP07]
- ▶ Initialization survey: [LMAC14]
- ▶ Beta divergence [Féld11]
- ▶ Non-negative tensor factorization [CiPh09]
- ▶ ...

pLSI [Hofm99] is Non-Negative Matrix Factorization [GaGo05]

Probabilistic Latent Semantic Indexing (c.f. Part 4) is a *probabilistic* version of LSI/LSA.

PLSI used a graphical mixture model with (mixture model notation; different than in Part 4)

$$P(w_i | d_j) = \sum_t P(t) P(d_j | t) P(w_i | t)$$

- ▶ W : word probability for each topic $P(w_i | t)P(t)$
- ▶ H : topic probability for each document $P(d_j | t)$
- ▶ probabilities are non-negative

“Any (local) maximum likelihood solution of PLSA is a solution of NMF with KL divergence.”
[GaGo05]

History of Non-negative Matrix Factorization

Popularized by Daniel D. Lee, and H. Sebastian Seung.

Learning the parts of objects by non-negative matrix factorization

In: *Nature* 401, 4, 1999. [DOI: 10.1038/44565](https://doi.org/10.1038/44565).

Similar approaches already found in:

- ▶ “Non-negative Rank Factorization”
M.W. Jeter, and W.C. Pye. **A note on nonnegative rank factorizations**
In: *Linear Algebra and its Applications* 38, 3, 1981. [DOI: 10.1016/0024-3795\(81\)90018-5](https://doi.org/10.1016/0024-3795(81)90018-5)
Ji-Cheng Chen. **The nonnegative rank factorizations of nonnegative matrices**
In: *Linear Algebra and its Applications* 62, 11, 1984. [DOI: 10.1016/0024-3795\(84\)90096-X](https://doi.org/10.1016/0024-3795(84)90096-X)
- ▶ “Positive Matrix Factorization”
Pentti Paatero, and Unto Tapper. **Positive matrix factorization: A non-negative factor**



model with optimal utilization of error estimates of data values

In: *Environmetrics* 5, 16, 1994. DOI: [10.1002/env.3170050203](https://doi.org/10.1002/env.3170050203)

Problems of NMF

Interesting theoretical analysis:

David L. Donoho, and Victoria Stodden.

When Does Non-Negative Matrix Factorization Give a Correct Decomposition into Parts?

In: *NIPS*, 8, 2003

Observations:

- ▶ decomposition is not orthogonal (in contrast to SVD, PCA)
- ▶ decomposition is not unique (not even when $X = WH$ is achieved; not only permutations and scaling of components)
- ▶ in particular, invariant regions are decomposed arbitrarily (stop words $\approx$ invariants of the data)
- ▶ 0 values are problematic with multiplicative update strategy

Need additional constraints, such as sparsity regularization [EgKo04; Hoyer04].

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Technische Universität Dortmund
Informatik VIII
AG Data Mining

Telefon: (+49) 231 755-2777
Send Email

Otto-Hahn-Straße 12
44227 Dortmund

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